

BIBEK RAM

Biomedical Engineering Researcher | Intelligent Medical Devices | Embedded & Edge AI

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RESEARCH PROFILE

Biomedical engineering researcher and final-year B.Tech. candidate at SRM Institute of Science and Technology (GPA 9.78/10.0), working across medical-device research and development, embedded firmware, and applied AI for healthcare. Author of two peer-reviewed IEEE conference papers spanning privacy-preserving edge AI and multimodal biomedical sensing. Current work centers on patient-monitoring systems, embedded firmware architecture, device-communication protocols, and ECG signal workflows within a regulated medical-device R&D environment. Research interests focus on translating biomedical signals, multimodal sensing, and embedded/edge computing into deployable, clinically relevant healthcare technologies.

RESEARCH INTERESTS

Intelligent Medical Devices • Embedded & Edge AI for Healthcare • Biomedical Signal Processing • Multimodal Biomedical Sensing • Medical Imaging & AI • Embedded Biomedical Systems • Neuroengineering • Translational Healthcare Technology

EDUCATION

B.Tech. in Biomedical Engineering — SRM Institute of Science and Technology, Chennai, India

2022 – 2026

GPA: 9.78/10.0 • First Class with Distinction • Performance-Based Scholarship, Academic Year 2024–2025

RESEARCH PUBLICATIONS

A Universal Plug-and-Play Edge-AI Platform for Privacy-Preserving Multimodal Healthcare Inference

2026 *International Conference on Emerging Trends in Information, Communication & Systems (ICETICS)*, IEEE, July 2026

DOI: [10.1109/ICETICS69505.2026.11651080](https://doi.org/10.1109/ICETICS69505.2026.11651080)

Multimodal Emotion Recognition via Correlation of EEG, Thermal, and Digital Image Data

IEEE *International Conference on Communication, Signal Processing and Computing (ICCSP)*, 2025

DOI: [10.1109/ICCSP64183.2025.11088814](https://doi.org/10.1109/ICCSP64183.2025.11088814)

RESEARCH EXPERIENCE

Research Assistant / Student Researcher — Department of Biomedical Engineering, SRM Institute of Science and Technology, Chennai, India

During B.Tech., 2022 – 2026

Under Dr. U. Snekhalatha, Professor & Head, Department of Biomedical Engineering

- Conducted end-to-end interdisciplinary research in medical imaging, biomedical signal processing, and multimodal biomedical systems, spanning data acquisition, preprocessing, feature and modality integration, computational model development, system prototyping, experimental validation, and performance evaluation.
- Developed a multimodal emotion-recognition system integrating EEG signals, thermal imaging, and digital facial images; the research resulted in an IEEE conference publication on cross-modal data correlation for emotion recognition.
- Designed and developed the Universal Plug-and-Play Edge-AI Platform for Privacy-Preserving Multimodal Healthcare Inference, a modular architecture for deploying heterogeneous healthcare AI models on embedded hardware; the work led to a second IEEE conference publication and received the Best Project Award.

Under Dr. Ashwin Narashimhan, Professor, Department of Biomedical Engineering

- Independently designed and fabricated a probe sonicator prototype using SolidWorks and electrical design skills for precision mechanical design and engineering drawings; developed and tested the prototype by adapting an existing ultrasonic bath sonicator circuit and experimentally validated its operational functionality.
- Contributed to the design and development of a piezoelectric micropump for insulin delivery, exploring piezoelectric actuation for controlled fluid delivery in a biomedical drug-delivery application.

PROFESSIONAL EXPERIENCE

Biomedical R&D Engineer — Invitronics Medtech Pvt. Ltd., Pune, India

2026 – Present

- **Medical Device R&D & Prototyping:** Drove the research, design, and integration of a patient-monitoring system, applying requirements-driven engineering from early-stage prototyping through system-level verification.
- **Embedded Systems & Signal Processing:** Developed and debugged embedded firmware and device-communication pipelines, executing biomedical signal processing in alignment with rigorous industry research methodologies.
- **Regulatory Integration & Risk Management:** Embedded compliance frameworks directly into the R&D lifecycle, applying ISO 14971 (Risk Management) and IEC 60601 (Safety & Essential Performance) to hardware and system architecture.
- **Software Lifecycle & Quality Management:** Aligned software architecture and testing with IEC 62304, preparing rigorous documentation under ISO 13485 (QMS) and CDSCO Medical Device Rules to support future clinical deployment.

Regulatory and Technical Intern — Bureau of Indian Standards (BIS), Medical Equipment & Hospital Planning Department, New Delhi, India

June – July 2025

- **Pre-Standardization Research & Analysis:** Conducted systematic literature reviews and comparative technical evaluations of market-available dialyzer reprocessing machines to establish baseline performance metrics for a new national standard.
- **Standards Formulation & Risk Assessment:** Analyzed structural safety, material construction, and quality-verification protocols, applying ISO 13485 (Quality Management Systems), ISO 14971 (Risk Management), and CDSCO regulations to draft comprehensive device evaluation frameworks.
- **Clinical Benchmarking & Field Research:** Conducted technical benchmarking and clinical workflow assessments through on-site hospital visits and stakeholder consultations with industry manufacturers, including Atlantic Biomedical and Greyfalcon Healthcare, to validate real-world reprocessing constraints.
- **Key techniques applied:** systematic technical review, clinical workflow analysis, regulatory technical writing, hazard identification, and comparative market evaluation.

Lab Experience — Department of Biomedical Engineering, SRM Institute of Science and Technology, Chennai, India

During B.Tech.

- **Hardware Engineering:** Medical instrumentation design, PCB integration, embedded electronics, and microcontroller programming.
- **Computational Analysis:** Biomedical signal processing and medical image processing.
- **Mechanical Design:** 3D CAD (SolidWorks) and biomechanical modeling / segmentation.

SELECTED RESEARCH & ENGINEERING PROJECTS

Universal Plug-and-Play Edge-AI Healthcare Platform | Raspberry Pi 4 · TensorFlow Lite · Flask

- Designed an edge-AI platform for heterogeneous healthcare models with modular input handling, preprocessing, model management, inference, and post-processing.
- Explored practical local deployment for privacy-preserving, low-latency multimodal healthcare inference.

Multimodal Emotion Recognition System | EEG · Thermal Imaging · Facial Imagery

- Built a multimodal classification pipeline combining physiological and imaging modalities; the underlying research was presented at IEEE ICCSP 2025.

Multimodal Tuberculosis Screening System | Cough Audio · Chest X-ray · Clinical Data

- Developing a multimodal screening pipeline combining acoustic analysis, medical-image classification, and structured clinical information, with a pathway toward edge deployment for low-resource settings.

AI-Driven Closed-Loop Insulin Delivery System | MATLAB/Simulink · PID · Physiological Modeling

- Modeled glucose–insulin dynamics and implemented PID-based closed-loop dosing logic in simulation, linking physiological modeling, control systems, and medical-device design.

Smart Anti-Tremor Glove for Neuro-Rehabilitation | Flex Sensors · IMU · ML · Vibrotactile Feedback

- Developed a wearable concept integrating tremor sensing, machine-learning-based classification, closed-loop vibrotactile feedback, and remote-monitoring capability.

TECHNICAL SKILLS

Embedded Systems & Firmware: C, embedded systems, ARM Cortex-M concepts, UART/serial communication, frame extraction and validation, circular/ring/double buffering, DMA, interrupts, packet/command dispatch, state machines, watchdogs, embedded debugging, TouchGFX

AI / Machine Learning: Python, TensorFlow, TensorFlow Lite, scikit-learn, OpenCV, multimodal learning, edge inference, feature extraction, classification, model deployment

Biomedical Engineering: Patient monitoring, ECG/EEG, biomedical instrumentation, biosensors, physiological signal acquisition, waveform analysis, medical-device prototyping

Signal & Image Processing: MATLAB, Simulink, PSD analysis, biomedical signal processing, medical imaging, image preprocessing, computer vision, Librosa/acoustic analysis

Hardware & Prototyping: Raspberry Pi 4, Arduino, custom PCB design, sensor integration, IMU, flex sensors, IoT healthcare prototypes, hardware–software integration

Regulatory & Quality: ISO 13485, ISO 14971, CDSCO MDR 2017, FDA QSR/QMS concepts, risk assessment, technical documentation, medical-device standardization

Engineering Tools: Git/GitHub, Jupyter Notebook, Google Colab, Flask, Arduino IDE, SolidWorks, AutoCAD, 3D and medical segmentation workflows

HONORS, AWARDS & LEADERSHIP

- **Winner – Best Biomedical Engineering Project, BioEdge Expo 2026** — multimodal edge-AI healthcare platform prototype, SRMIST.
- **Research Presenter, IEEE ICCSP 2025** — presented peer-reviewed multimodal emotion-recognition research.
- **Performance-Based Scholarship**, Department of Biomedical Engineering, SRMIST.
- **Performance-Based Scholarship, High School**, Department of Higher Education, St. Xavier's College.
- **International Student Ambassador**, SRMIST International Relations — mentored fellow students on biomedical research projects, real-time inference applications, and engineering design thinking.
- **Student Ambassador, TekMedica Club, SRMIST** — supported biomedical innovation activities and technical workshops.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Biomedical AI, Machine Learning & Programming

Complete Machine Learning A-Z™: 10 Real-World Projects — Udemy (MG Analytics), Mar 2025 · Machine Learning Using Python — NIELIT Calicut, 2024 · Programming for Everybody (Getting Started with Python) — University of Michigan via Coursera, Apr 2024 · MATLAB Onramp — MathWorks, Jul 2023

Neuroengineering & Neuroscience

Computational Neuroscience — University of Washington via Coursera, Jul 2023 · Fundamentals of Neuroscience for Neuroimaging — Johns Hopkins University via Coursera, Sep 2024

Healthcare, Medical Devices & Regulatory Affairs

Introduction to Healthcare — Stanford University via Coursera, Apr 2024 · Master Medical Device Regulatory Affairs — Udemy (Access Health Academy), Jan 2026 · Medical Device Fundamentals — SkillMed, Aug 2026 · Indian MDR (Medical

Devices Rules, 2017) — SkillMed, Aug 2026 · Fundamentals of Good Documentation Practices (GDP) for Medical Devices — SkillMed, Aug 2026

Nanotechnology & Sensors

Nanotechnology: A Maker's Course — Duke University / NC State University / UNC Chapel Hill via Coursera, Aug 2023 · Nanotechnology and Nanosensors, Part 2 — Technion – Israel Institute of Technology via Coursera, Feb 2024

Credentials Verification: [Certificates](#)

REFERENCES

Dr. U. Snehalatha

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